

No.of Functional Features Included in the Solution

Date	1 July , 2026
Team ID	
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview

This document lists the functional features implemented in the **OptiCrop – Smart Agricultural Production Optimization Engine**. The application uses Machine Learning and a Flask web application to recommend the most suitable crop based on soil nutrients and environmental conditions.

Functional Features

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Soil Data Input	Accepts soil nutrient and environmental parameters through a web form.	Flask UI	Done	High
2	Data Preprocessing	Processes user input and prepares data for prediction.	Data Processing	Done	High
3	Crop Recommendation	Predicts the most suitable crop using the trained Machine Learning model.	Prediction Engine	Done	High
4	Model Integration	Loads the trained Random Forest model using Joblib for prediction.	Backend	Done	Medium
5	Crop Information Display	Displays crop image, growing season, temperature, water requirement, and farming tips.	Frontend	Done	Medium
6	Responsive Web Interface	Provides an easy-to-use and responsive interface for users.	Frontend	Done	Medium
7	Error Handling	Validates user input and prevents invalid predictions.	Backend	Done	Medium
8	Model Deployment	Deploys the application using Flask and Render Cloud Platform.	Deployment	Done	Medium

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Soil Data Input, Data Preprocessing, Crop Recommendation, Model Integration
Additional Features	Crop Information Display, Responsive UI, Error Handling, Cloud Deployment
Features Tested & Verified	8 Features Successfully Tested

Feature Category Breakdown

S.No	Category	Features in Category	Example Feature
1	User Interface (UI)	3	Home Page, Crop Recommendation Form, Result Page
2	Backend / Logic	3	Data Preprocessing, Prediction Engine, Error Handling
3	Machine Learning	1	Random Forest Crop Prediction Model
4	Integration	1	Flask Integration with Machine Learning Model
5	Deployment	1	Render Cloud Deployment